

Exhibit CC-120

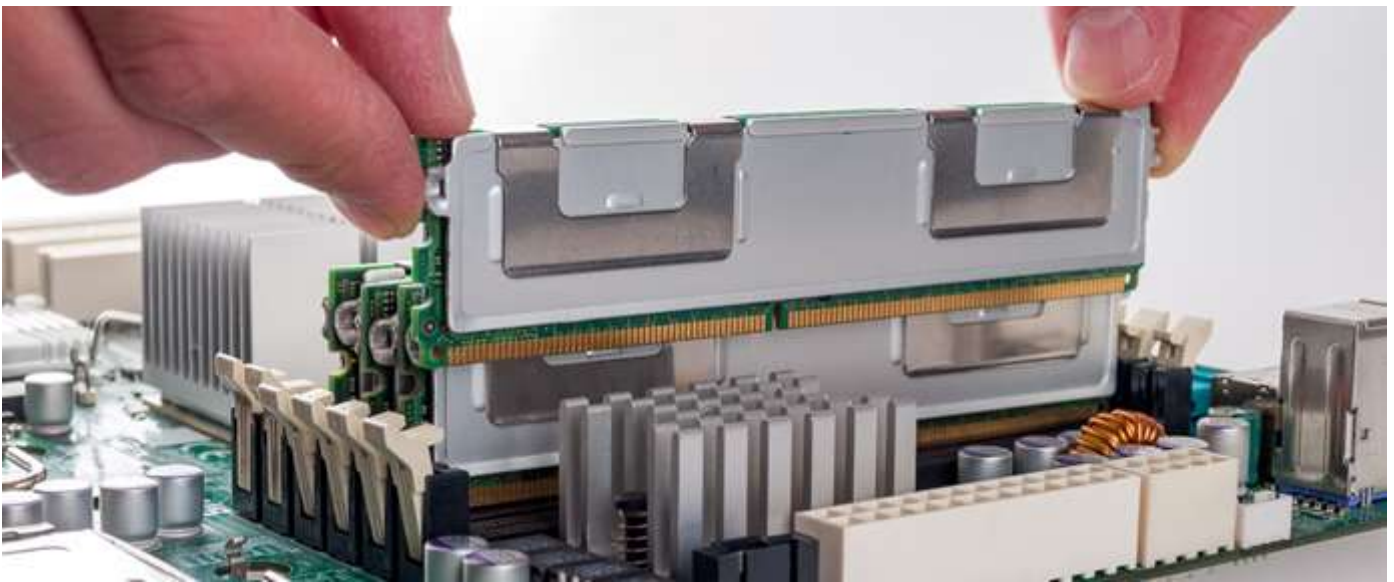


Tech - Article 2022.03.29

Memory VR changes in the system by DDR5 adoption



▶0805i, 10uF, 25V, X6S (CL21X106KAYQNWE)



Abstract

Because of DDR5 memory module with built-in VR, power line connected to DIMM socket in server or network system needs to be changed to 12V. Because of this, 25V MLCC is now required, which Samsung Electro-Mechanics offers a 0805i, 10uF, 25V MLCC in X5R (85°C) and X6S (105°C) dielectrics.

DDR5 - 2X faster memory than DDR4

After 7 years since DDR4 was released in 2014, memory makers started to introduce DDR5. DDR5 supports up to twice the memory speed than DDR4 and memory voltage is lowered as 1.1V is better for power efficiency. This is the important aspect from a system designer's point of view, because memory VR needs to be close with DRAM in

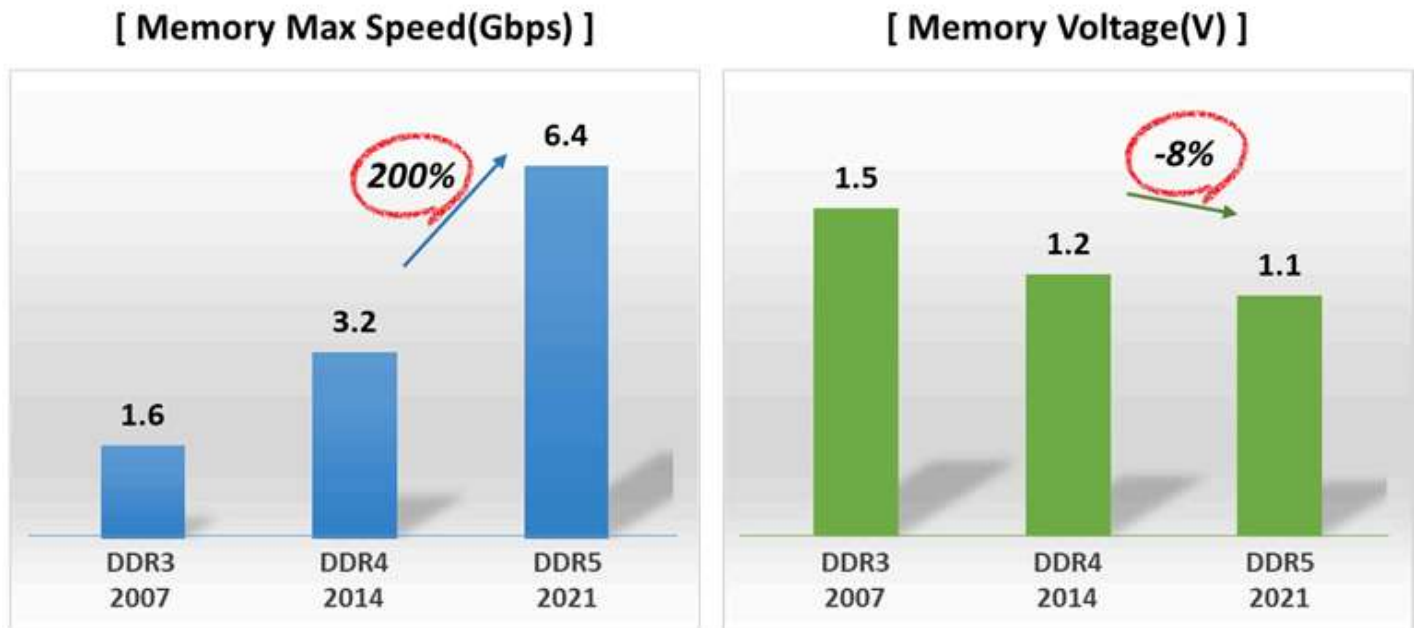


Figure1. DDR4 vs DDR5 JEDEC standard spec.

Need to support 12V power line inside motherboard

Voltage converter for memory was located in the system and converts 12V to 1.2V and delivers power to DIMM socket directly. But 12V power line is directly connected to DIMM socket from DDR5 and built in VR converts to 1.1V, which delivers power to DRAM. As VR disappears from system, the system design has the advantage of being simplified, but on the contrary, the memory module need a lot of components. As such, the number of various components such as PMIC, Power Inductor, and MLCC will increase. From a server and network system designer's point of view, power line voltage from PSU to DIMM socket will be increased from 1.2V to 12V, so the system needs new 25V MLCC which need to be located as close to the DIMM socket as possible.



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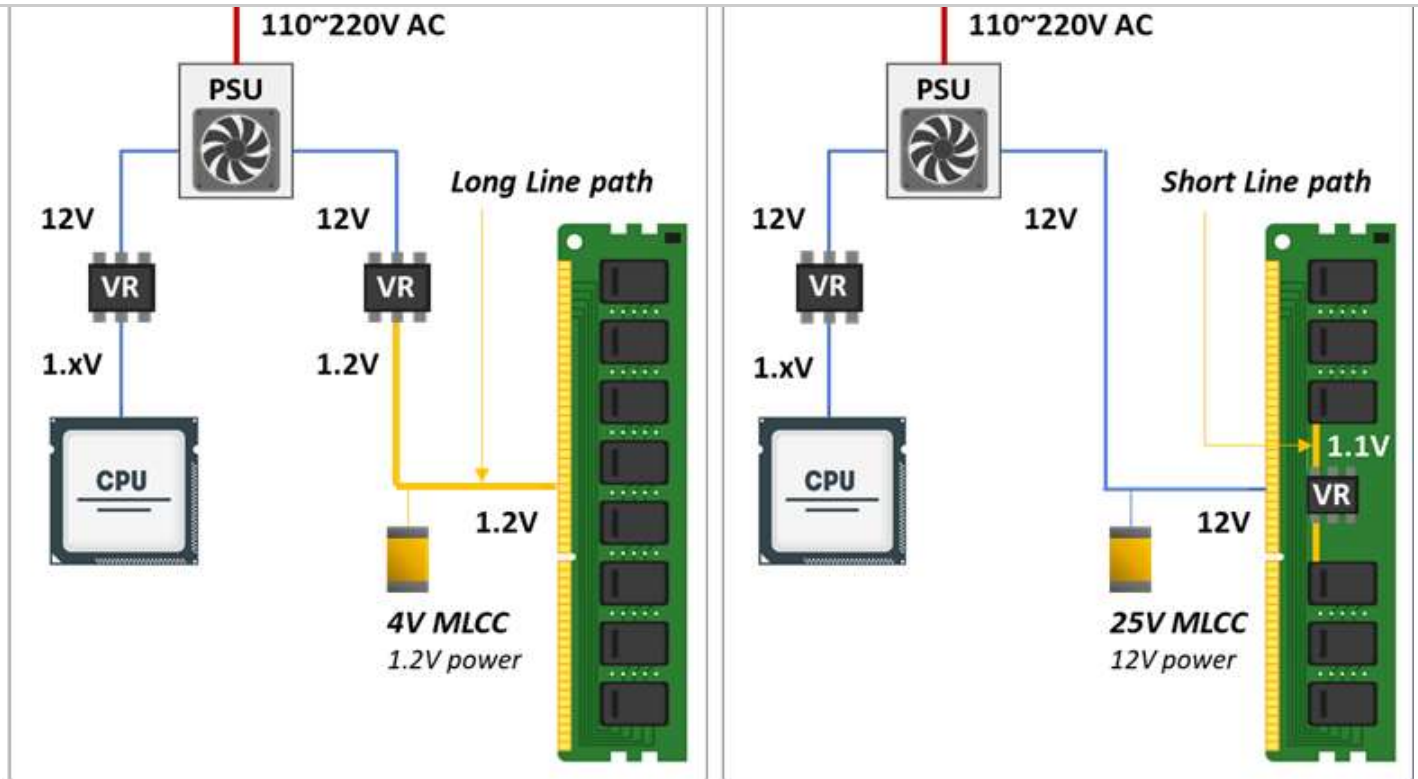


Figure2. DDR4 Based-system vs DDR5 Based-system

Thermal issue from DDR5 module

An additional factor for MLCC selection is temperature. The area where DRAM is located was regarded as a cool zone because DDR4 memory module didn't generate a lot of heat. Unlike DDR4, DDR5 is now to be considered a heat zone due to the newly added PMIC. Also, due to the increase in the number of DIMMs, as the density increases, the temperature continues going up.

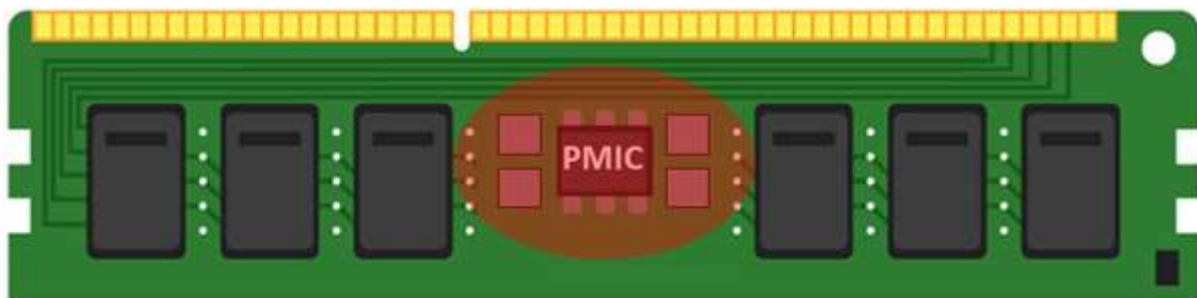


Figure4. DDR5 memory module layout

Due to this, 105°C X6S level or higher temperature rated MLCC is required.

Samsung 0805i, 10uF, 25V, X6S Solution (CL21X106KAYQNWE)



mechanics developed a 0805i, 10uF, 25V, X6S MLCC (CL21X100KATQNNWE), which are available for purchase through our partners Digi-Key and Mouser Electronics.

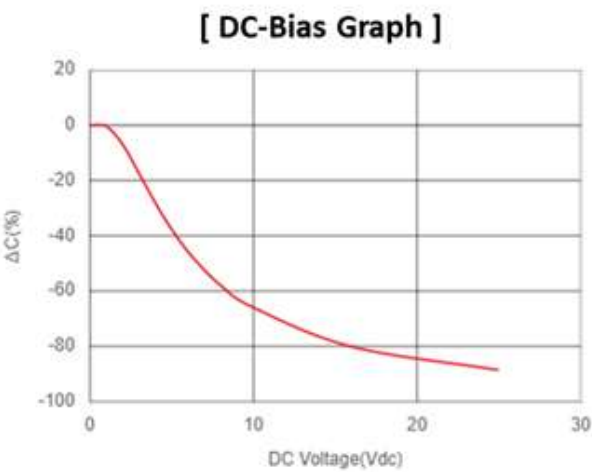


Figure5. CL21X106KAYQNNWE DC-Bias

[Reliability Condition]

	General	High Reliability *
Biased Humidity	40°C 95%RH	65°C 90%RH
High Temp. Load Test	Max Temp., 1.0Vr, 1000hr	Max Temp., 1.5Vr, 1000hr

Figure6. CL21X106KAYQNNWE reliability spec.

Datasheet Download

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Samsung 0805i, 22uF, 25V, X5R Solution (CL21A226MAYNNWE)

If X6S isn't needed, Samsung offers an X5R variant at a higher capacitance value(22uF) with same 0805i case size as well.

Datasheet Download

Buy at Mouser Electronics

If you have an additional inquiry including sample, Please click [here](#).

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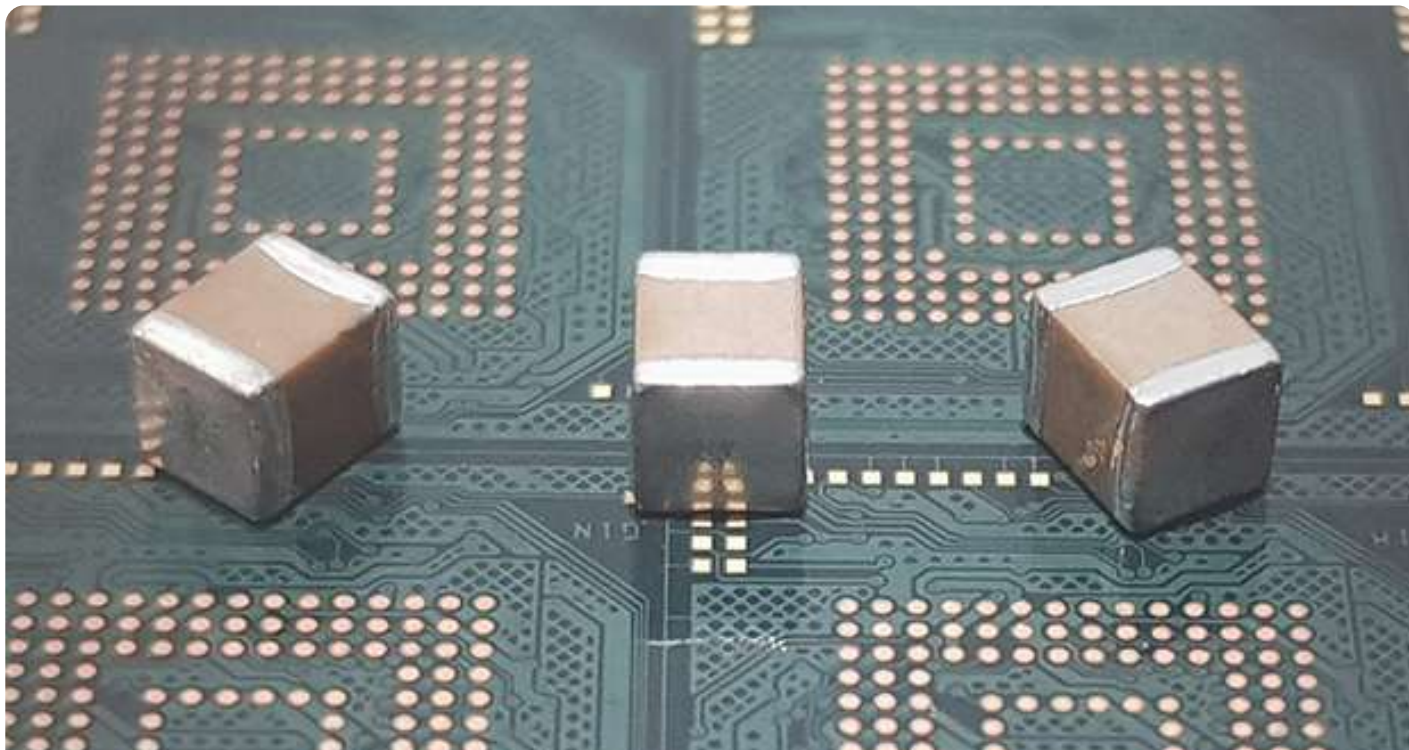
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